



FNB SALES *Performance*



Objective

The objective was to calculate these metrics and use them to understand the product's sales and profitability performance. These are the 5 main questions I worked on:

1. Daily sales price per unit
2. Average unit sales price
3. Daily gross profit %
4. Gross profit % per unit
5. Price elasticity during 3 promotional periods





Data Overview

The dataset contains daily trading information for one product. I used SQL in Databricks to clean, transform and calculate the required metrics.

For each daily record on the products, they all had the;

1. Date
2. Sales
3. Cost of Sales
4. Quantity Sold
5. Number of records/days

Daily Sales Unit Pricing

Total Sales ÷ Quantity
Sold = Price per Unit



Average Selling Price≈ R37.07

ROUND(AVG(Sales / Quantity_Sold), 2)
AS avg_unit_price_simple_mean

Weighted Average Unit Price ≈ R35.40

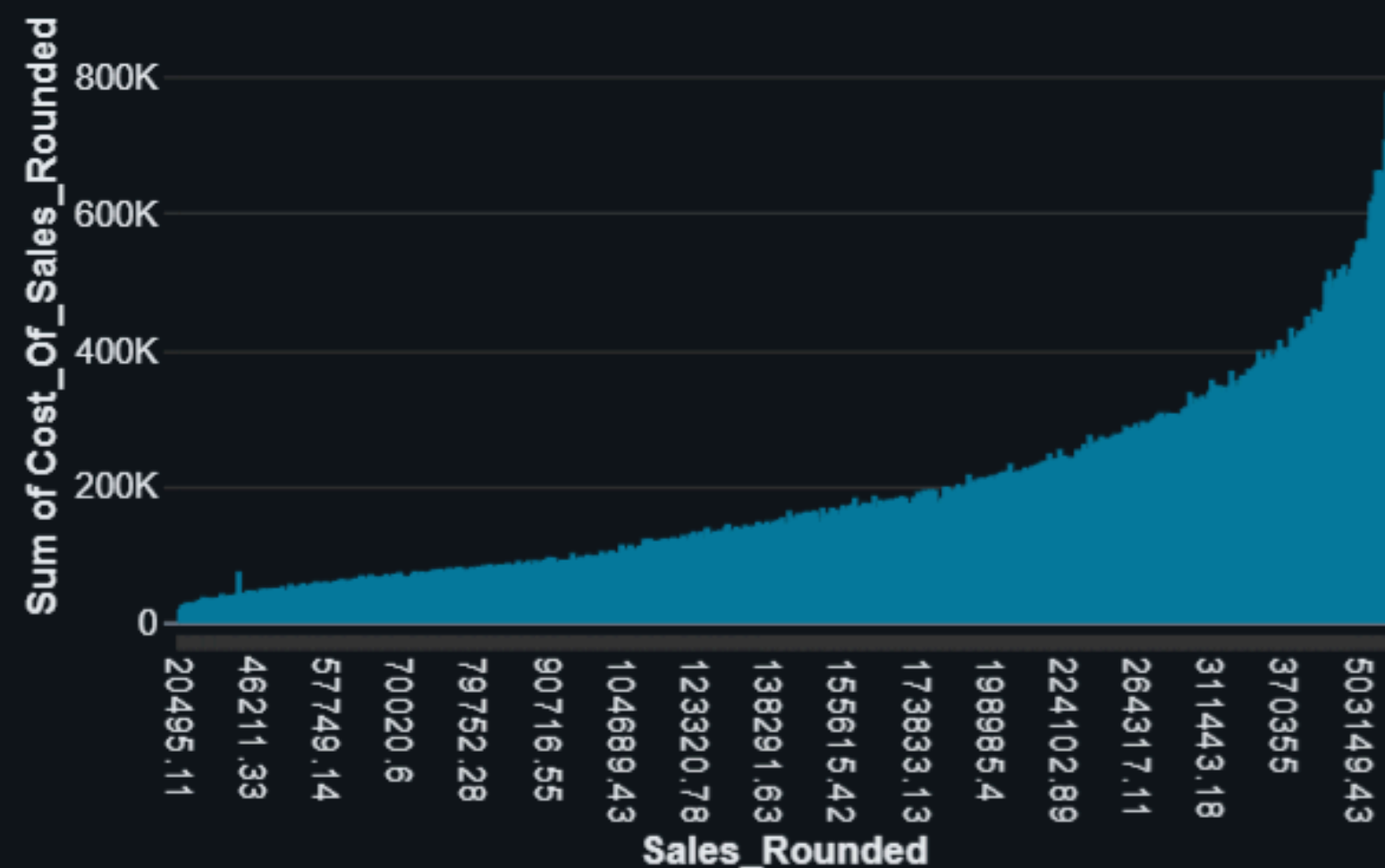
ROUND(SUM(Sales) / SUM(Quantity_Sold),
2) AS avg_unit_price_volume_weighted



Profit/Loss

KEY PERFORMANCE INDICATORS

Cost of Sales by Sales



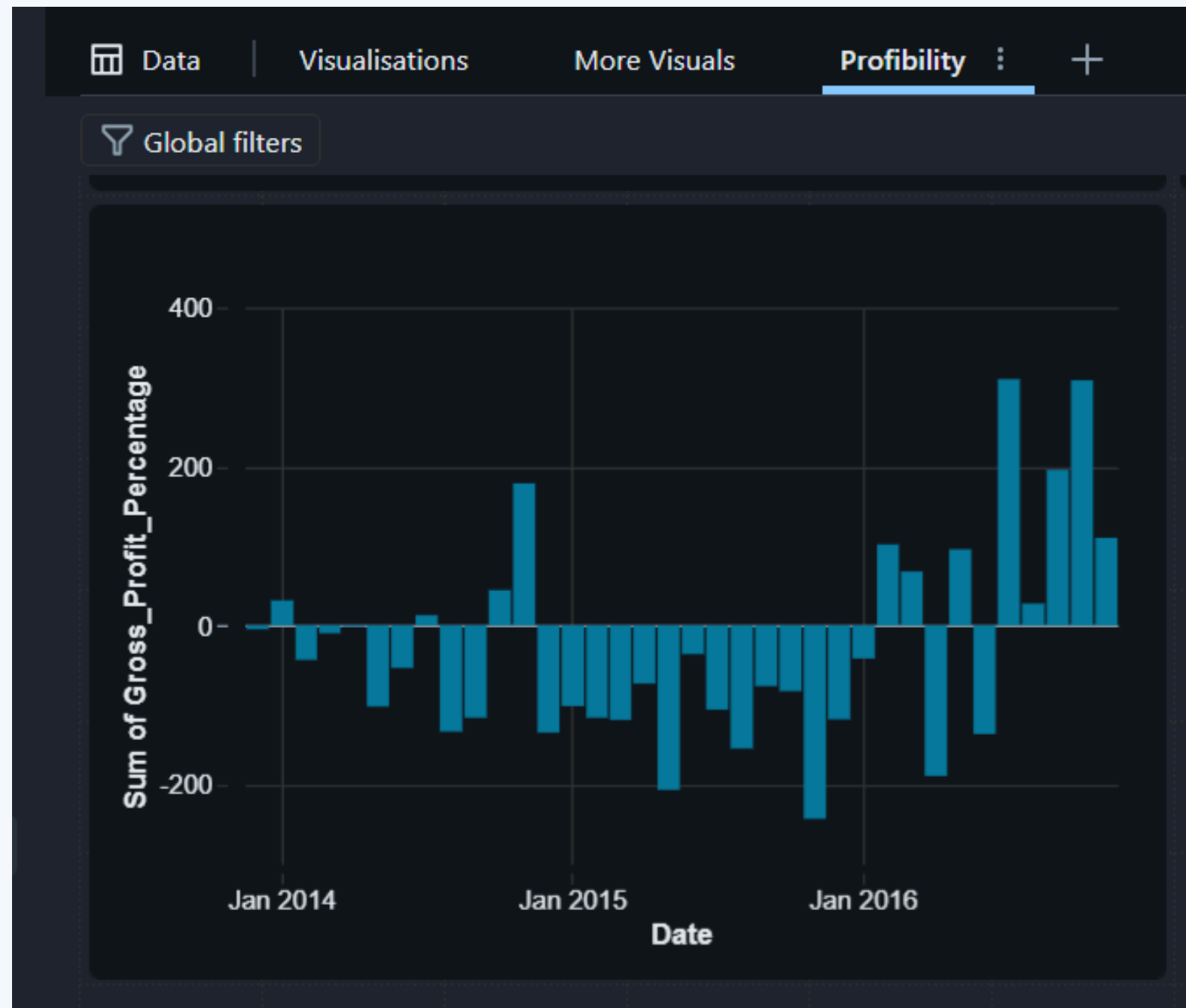
- Total Sales: $\approx$ R186,9091,828.12M
- Total Cost: $\approx$ R194.03M
- Gross Profit: $\approx$ -R7.12M
- Overall Gross Margin: $\approx$ -3.81%

The main profitability finding is that total cost of sales exceeded total sales, resulting in an overall gross loss.



Daily Gross Profit %

Date By Gross Profit %



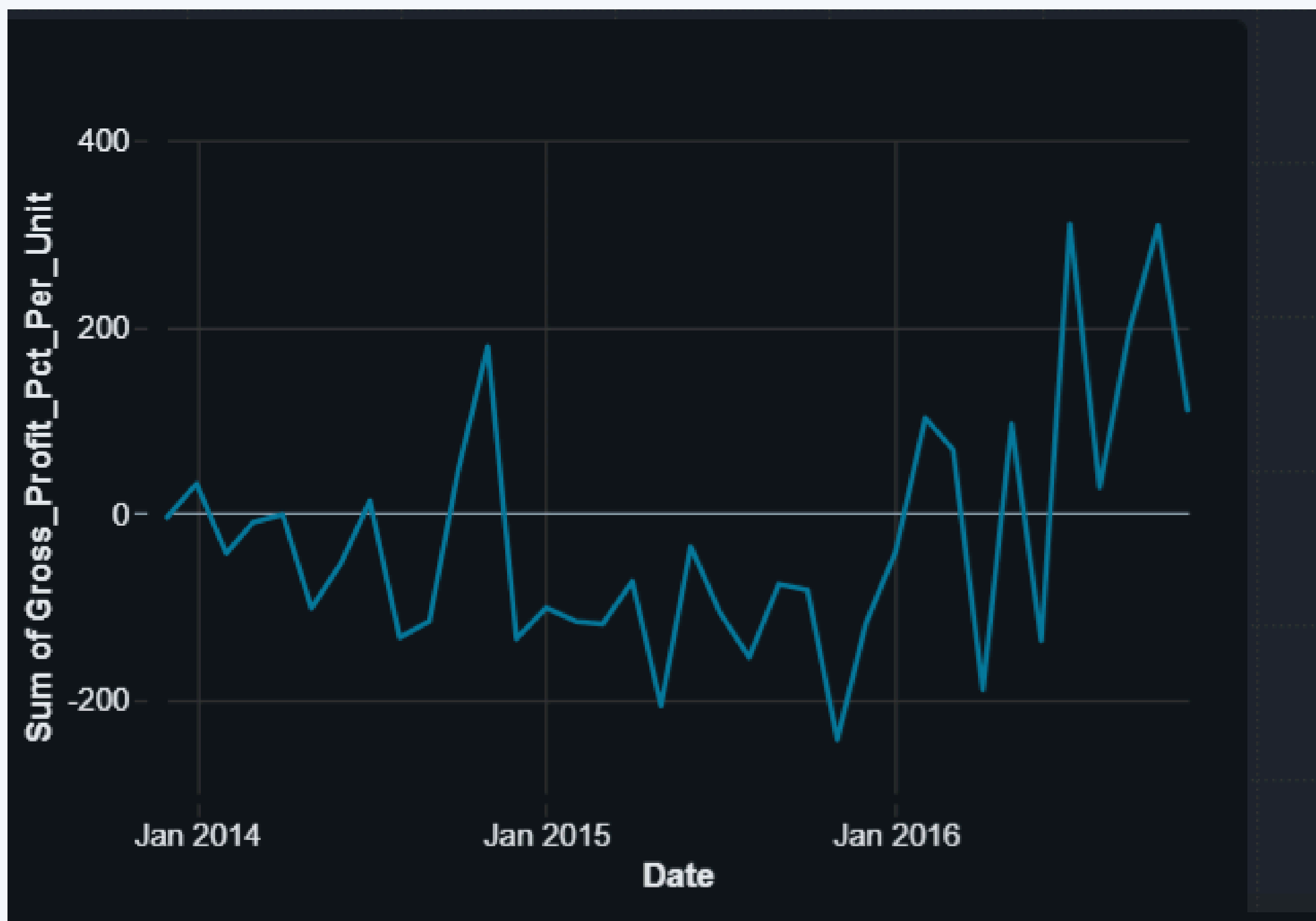
From the January 2014 the business made 31.45% worth of profit. In these period, the sales were still high. Beginning of Feb-July the sales dropped. In this case the Profitability varies significantly from day to day.

In Databricks we calculate by;

```
ROUND((Sales - Cost_Of_Sales) / Sales * 100, 2) AS  
Gross_Profit_Percentage
```



Gross Profit % Per Unit



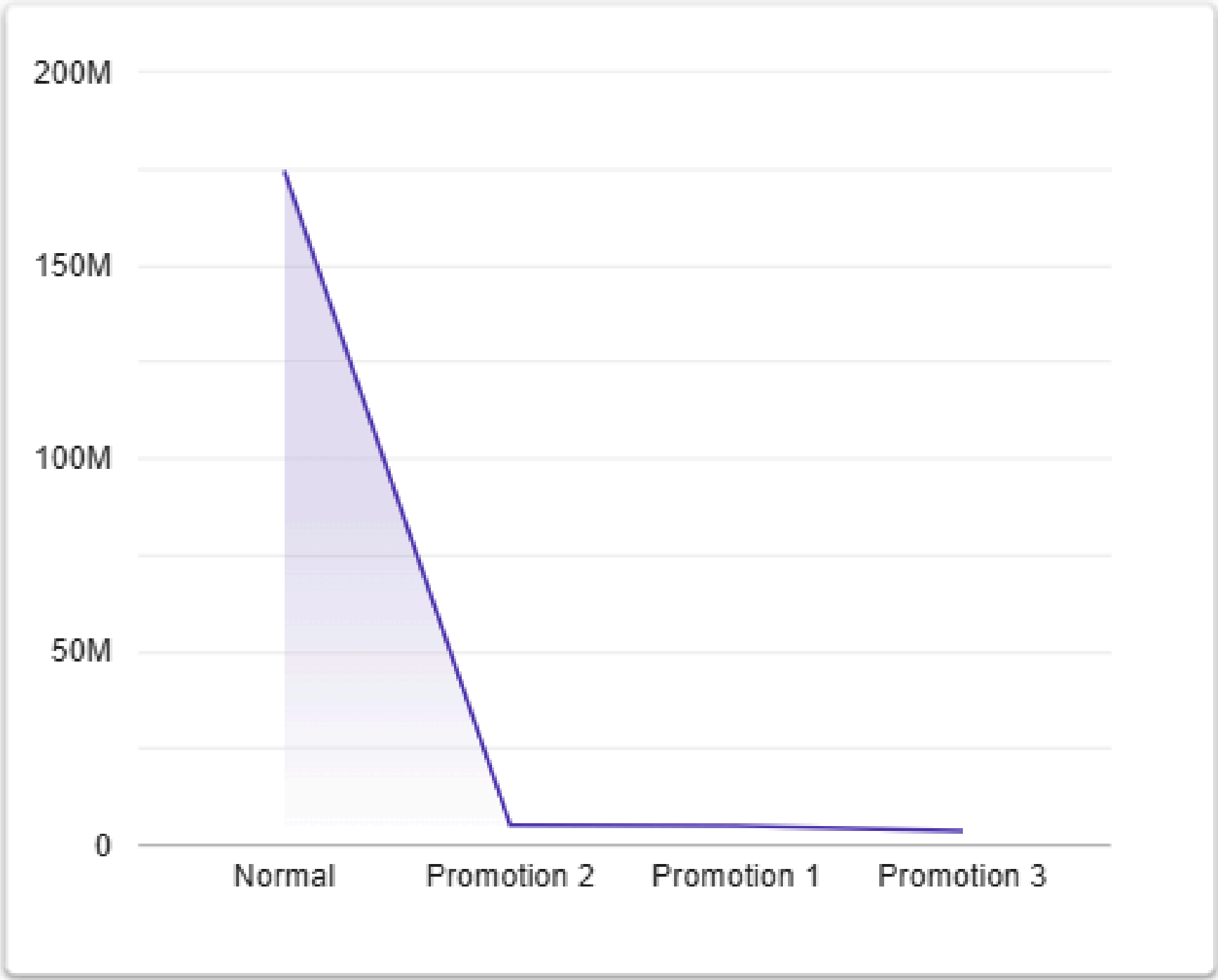
This metric shows how much profit or loss was generated from each unit sold on different dates

In Databricks we calculate by;
`ROUND(Sales - Cost_Of_Sales, 2) AS Profit`



Elasticity Demand

Profit By Period_Type By Total Sales



	Period_Type	Profit ▾
1.	Promotion 3	-207,463.18
2.	Promotion 2	-392,670.35
3.	Promotion 1	-413,162.61
4.	Normal	-6,102,985.06

Key Insights

- The average selling price $\approx$ R35.40.
- High promotional volume did not make any profit rather it increased more costs.
- Promotional periods brought good stock turn over in terms of sales.
- However, profitability needs to remain a key consideration when evaluating promotional pricing.



Recommendations

This product is losing money more often than not, and Promotions are making it worse, not better.

My recommendations:

- 1. Pause or rework the current promotional pricing model for this product —
Investigate loss-making periods*
- 2. Revisit the cost base or supplier pricing, because even "normal" days are barely breaking even.*
- 3. Test promotions against profitability, not only quantity*





Thank *You*



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